NAME: ____________________ **PERIOD:** ________ **DATE:** ____________

Two Samples of Service Times

AP Statistics · Topic 7.6 (CED 4.7) · B: 2027-01-25 · E: 2027-03-12

[STAR] TODAY'S PROBLEM

Suppose two depots independently take SRSs of 36 weekday service-time records from monthly populations of 900 North and 1,200 South records. Technology gives $df = 69.31$ and $t^* = 1.995$ for a 95% two-sample interval.

Tariq: "I sorted both samples and paired equal ranks. The resulting differences have $\bar{d} = 2.2$ and $s_d = 3.0$, giving $(1.18, 3.22)$. Pairing makes the estimate more precise."

Mei: "The records were sampled separately, so I would use a two-sample interval. Sorting observed responses may change the uncertainty rather than reveal design-based pairs."

Service-time samples (min)

Depot	N	n	$\bar{x}$	s
North	900	36	42.3	8.4
South	1,200	36	40.1	7.6

FIRST TAKE — before the video (5 min)

Which interval procedure matches the original design? Cite any link that did or did not exist before times were observed.

[BOOK] LET THE DESIGN CHOOSE THE INTERVAL (keep on the page)

For two independent random samples, use the two-sample t interval $(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{s_1^2/n_1 + s_2^2/n_2}$. Check independent random sampling or random assignment, each 10% condition when sampling without replacement, and an approximately normal sampling distribution (population shape or both samples over 30). Matched pairs must be created by the design before responses are observed; sorting two independent samples does not create pairs. Interpret the resulting interval in the stated subtraction order.

Finish after the video:

DOK 1 **(a)** Define $\mu_N - \mu_S$, calculate its point estimate, and name the design-based interval procedure.

DOK 2 **(b)** Verify random, 10 percent, and shape conditions. Compute SE, margin of error, and the 95% interval.

DOK 3 **(c)** Adjudicate the procedures using the separate SRSs, rank pairing, $s_d = 3.0$, and both intervals. Explain how response-based pairing changes uncertainty and the conclusion a reader might draw; then interpret the warranted interval and bound its population scope.

Frame: The design supports a ________________ interval because ______________________; sorting ranks instead ____________________.

Frame: The two uncertainty estimates are ________________ and ________________, which changes ____________________.

EXIT — turn this in

One thing the video changed about my first take: ________________________________